

WASP-104b

R-2042

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_260401 · created 2026-08-08T23:11:39+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461131.6744 +/- 0.0067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.044
Transit depth (Rp/Rs)^2	2.27 +/- 1.32 [%]
Orbital Inclination (inc)	87.39 +/- 2.93
Airmass coefficient 1 (a1)	3.58 +/- 1.73
Airmass coefficient 2 (a2)	-0.79 +/- 1.39
Scatter in the residuals of the lightcurve fit is	49.51 %
Best Comparison Star	None
Optimal Aperture	0.88
Optimal Annulus	3.8
Transit Duration (day)	0.074 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.044 vs 0.1214 (deviation 0.65σ)
Duration fit vs published	0.074 d vs 0.0742 d (deviation 0.01σ)
Mid-time O–C	14.7 min
Depth SNR	3.3
Residual scatter	49.51 %

Light curve

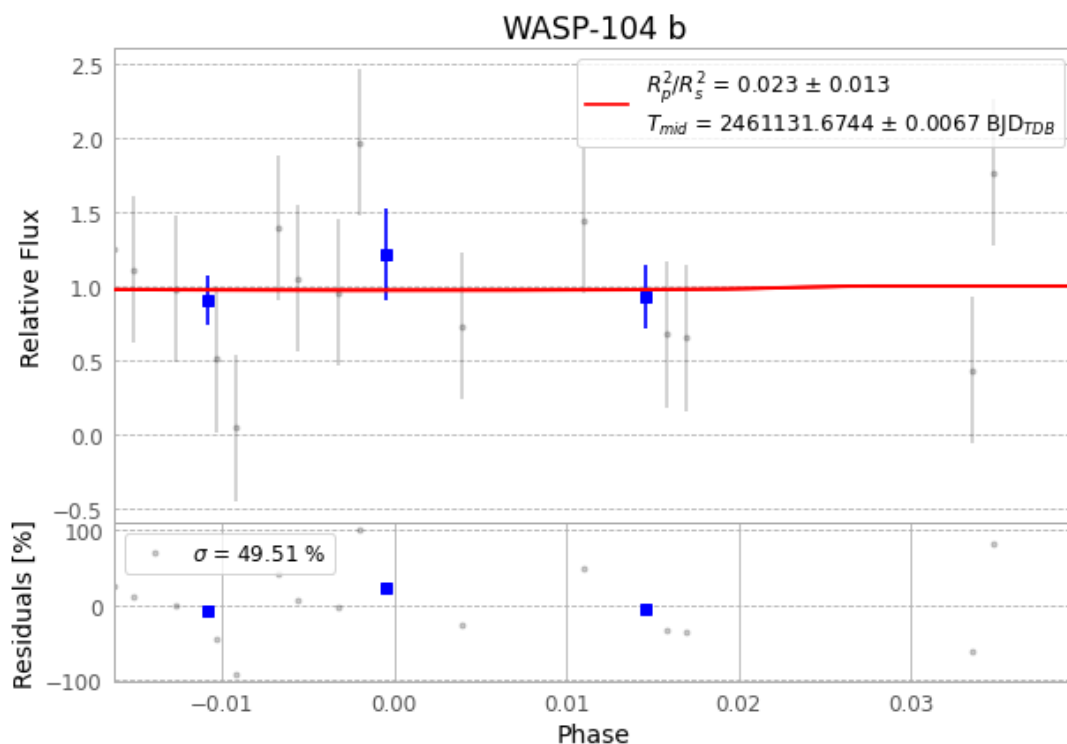


Figure 1 — FinalLightCurve_WASP-104 b_2026-03-31.png (SHA-256 3a365c26bf6dd713...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [46, 418], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 76c82b991a45b102...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), WASP-104260401030315.FITS → WASP-104260401061816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-260401.log (386fcabd505d...), FinalLightCurve_WASP-104 b_2026-03-31.png (3a365c26bf6d...), FinalLightCurve_WASP-104 b_2026-03-31.csv (3896ecf4c1d0...)

Compute resources

Wall time 375.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:11Z.